

CLOB and BLOB in MySQL

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CLOB stands for Character Large Object, in general, a SQL CLOB is a built-in data type which is used to store large amount of textual data. Using this datatype, you can store 4GB of data in MySQL Database.

The MySQL database provides support for CLOB datatype as TINYTEXT, TEXT, MEDIUMTEXT, LONGTEXT.

A **CLOB (Character Large Object)** is a SQL-standard data type designed to store very large text data—often gigabytes in size.

MySQL historically **does not provide a formal CLOB type**, but its TEXT types fulfill the same role.

MySQL Equivalent Types

MySQL's TEXT-family types act as CLOB equivalents:

MySQL Type Max Size Typical Use

TINYTEXT	255 bytes	Small text snippets
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TEXT	65 KB	Medium-length text
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MEDIUMTEXT	16 MB	Large text blocks
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LONGTEXT	4 GB	Very large documents, logs, JSON, XML
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These sizes align with typical CLOB use cases such as storing documents, logs, or structured text.

Inserting Data into a column of CLOB type

You can insert a CLOB type value using the `setCharacterStream()` or `setClob()` method of the `PreparedStatement` interface.

These methods accept two parameters -

- An integer holding the index of the bind variable (representing the column that stores data of type CLOB.)
- A reader object holding the file containing the data you need to store into the table.

```
FileReader reader = new FileReader("fileNameWithPath");
```

```
pst.setClob(1, reader);
```

Or

```
pst.setCharacterStream(1, reader);
```

A BLOB is a binary Large Object that can hold a variable amount of data with a maximum length of 65,535 characters.

These are used to store large amount of binary data such as images or other types of files. MySQL support BLOB data type using TINYBLOB, MEDIUMBLOB and LONGBLOB.

The `java.sql.BLOB` interface of the JDBC API represents the BLOB datatype. Since the BLOB object in the JDBC is implemented using a SQL locator, it hold a logical pointer the SQL BLOB(not the data).

Inserting data into a column of BLOB data type

You can insert block data into a table using the `setBinaryStream()` or `setBlob()` methods of the prepared statement interface. These methods accept two parameters.

- An integer holding the index of the bind variable (representing the column that stores the data of the type BLOB).
- An **InputStream** object holding the file containing the binary data (image or audio or video) you need to store into the table.

```
InputStream inputStream = new FileInputStream("fileNameWithPath");  
pstmt.setBlob(1, inputStream);
```

Or

```
Pstmt.setBinaryStream(1, inputStream);
```

Following JDBC program establishes connection with the Mysql and insert BLOB and CLOB values into the table `articles_data`.

The screenshot shows a database table creation interface. At the top, the table name is 'articles_data', the schema is 'carshowroom', the charset/collation is 'utf8mb4', and the engine is 'InnoDB'. Below this, there is a table with columns: ArticleID (INT, PK, NN), ArticleImage (BLOB), Article_Description (MEDIUMTEXT), and Price (DECIMAL(10,0)).

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
ArticleID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
ArticleImage	BLOB	☐	☐	☐	☐	☐	☐	☐	☐	NULL
Article_Description	MEDIUMTEXT	☐	☐	☐	☐	☐	☐	☐	☐	NULL
Price	DECIMAL(10,0)	☐	☐	☐	☐	☐	☐	☐	☐	NULL

Structure of the table `articles_data`.

WAP using JDBC to insert data in the above given table.